

02YMECH - Solution 13

Assigned for the week of Dec 15, 2025

Solution

(a) For torque-free motion, Euler's equations are

$$\begin{aligned}I_1\dot{\omega}_1 &= (I_2 - I_3)\omega_2\omega_3, \\I_2\dot{\omega}_2 &= (I_3 - I_1)\omega_3\omega_1, \\I_3\dot{\omega}_3 &= (I_1 - I_2)\omega_1\omega_2.\end{aligned}$$

The kinetic energy is

$$T = \frac{1}{2} (I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2).$$

Differentiating and using Euler's equations gives

$$\dot{T} = I_1\omega_1\dot{\omega}_1 + I_2\omega_2\dot{\omega}_2 + I_3\omega_3\dot{\omega}_3,$$

and substituting Euler's equations yields

$$\dot{T} = \omega_1(I_2 - I_3)\omega_2\omega_3 + \omega_2(I_3 - I_1)\omega_3\omega_1 + \omega_3(I_1 - I_2)\omega_1\omega_2 = 0,$$

$$\dot{T} = 0,$$

so T is conserved.

The angular momentum in the body frame is

$$\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3),$$

and its squared magnitude is

$$L^2 = I_1^2\omega_1^2 + I_2^2\omega_2^2 + I_3^2\omega_3^2.$$

Differentiating,

$$\dot{L}^2 = 2I_1^2\omega_1\dot{\omega}_1 + 2I_2^2\omega_2\dot{\omega}_2 + 2I_3^2\omega_3\dot{\omega}_3,$$

and substituting Euler's equations yields

$$\dot{L}^2 = 2I_1\omega_1(I_2 - I_3)\omega_2\omega_3 + 2I_2\omega_2(I_3 - I_1)\omega_3\omega_1 + 2I_3\omega_3(I_1 - I_2)\omega_1\omega_2 = 0,$$

satisfies $\dot{L}^2 = 0$. Hence L^2 is conserved. *Even though $\mathbf{L}$ itself is not conserved in the body frame, its magnitude is.*

(b) From the initial conditions,

$$T = \frac{1}{2}(I_1 + I_2)\omega_0^2, \quad L^2 = (I_1^2 + I_2^2)\omega_0^2.$$

Thus, for all t ,

$$I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2 = (I_1 + I_2)\omega_0^2, \quad (1)$$

$$I_1^2\omega_1^2 + I_2^2\omega_2^2 + I_3^2\omega_3^2 = (I_1^2 + I_2^2)\omega_0^2. \quad (2)$$

Solving (1)–(2) for ω_2^2 and ω_3^2 yields

$$\omega_3^2 = \frac{I_2 - I_1}{I_3 - I_1} (\omega_1^2 - \omega_0^2), \quad \omega_2^2 = \frac{I_1(I_3 - I_2)}{I_2(I_2 - I_1)} (\omega_0^2 - \omega_1^2).$$

From the first Euler equation,

$$\dot{\omega}_1 = \frac{I_2 - I_3}{I_1} \omega_2 \omega_3.$$

Squaring and substituting the expressions above gives

$$\dot{\omega}_1^2 = \frac{(I_3 - I_2)(I_2 - I_1)}{I_1^2(I_3 - I_1)} \omega_1^2 (\omega_0^2 - \omega_1^2).$$

(c) Since $\dot{\omega}_1^2 \geq 0$,

$$0 \leq \omega_1^2 \leq \omega_0^2, \quad \text{or} \quad -\omega_0 \leq \omega_1 \leq \omega_0.$$

The motion is bounded between turning points and therefore periodic.

(d) Let

$$C = \frac{(I_3 - I_2)(I_2 - I_1)}{I_1^2(I_3 - I_1)}.$$

Then

$$\frac{d\omega_1}{dt} = \sqrt{C} \omega_1 \sqrt{\omega_0^2 - \omega_1^2}.$$

Separating variables,

$$dt = \frac{d\omega_1}{\sqrt{C} \omega_1 \sqrt{\omega_0^2 - \omega_1^2}}.$$

One quarter of the period corresponds to $\omega_1 : 0 \rightarrow \omega_0$:

$$\frac{T}{4} = \frac{1}{\sqrt{C}} \int_0^{\omega_0} \frac{d\omega_1}{\omega_1 \sqrt{\omega_0^2 - \omega_1^2}}.$$

With the substitution $\omega_1 = \omega_0 \sin \theta$,

$$T = \frac{4}{\omega_0 \sqrt{C}} \int_0^{\pi/2} \frac{d\theta}{\sin \theta},$$

which is an integral representation of the motion.